

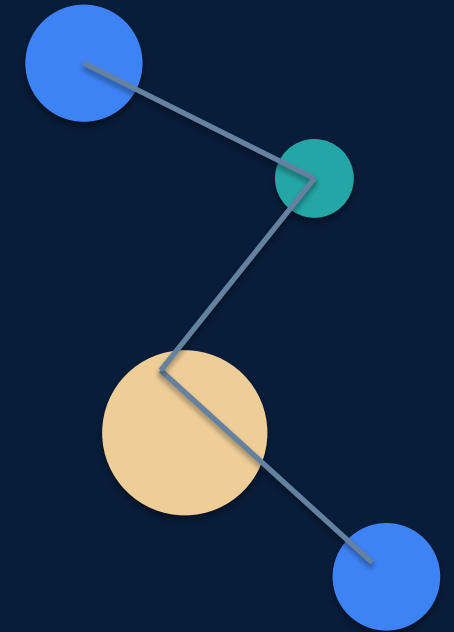
WEB & INTERNET

FUNDAMENTALS

From the Internet and the Web to development tools, servers, pages and HTML

LECTURE 02

Frontend Development • HTML, CSS & JavaScript



What this lecture covers

A connected view of how the Web works and the tools used to build for it.

01

Internet & Web

Networks, ARPANET, Internet vs Web

02

Protocols

HTTP, HTTPS, SMTP and FTP

03

URL Journey

DNS, IP, request, response and rendering

04

Development Tools

Node.js, package managers and VS Code

05

Web Foundations

Servers, websites, pages, folders and page types

06

HTML Evolution

From early HTML to the Living Standard

Network → Internet → Web

Three related ideas, but they are not the same thing.

Computer Network

A computer network is a group of connected devices that can exchange data and share resources. A home Wi-Fi network is a simple example.

Internet

The Internet is a global system of interconnected networks. It provides the infrastructure that allows devices and networks around the world to communicate using Internet protocols.

World Wide Web

The Web is an information system that runs on top of the Internet. It connects web pages and other resources through URLs and hyperlinks, typically accessed with browsers using HTTP or HTTPS.

NETWORK

Local connectivity

INTERNET

Global infrastructure

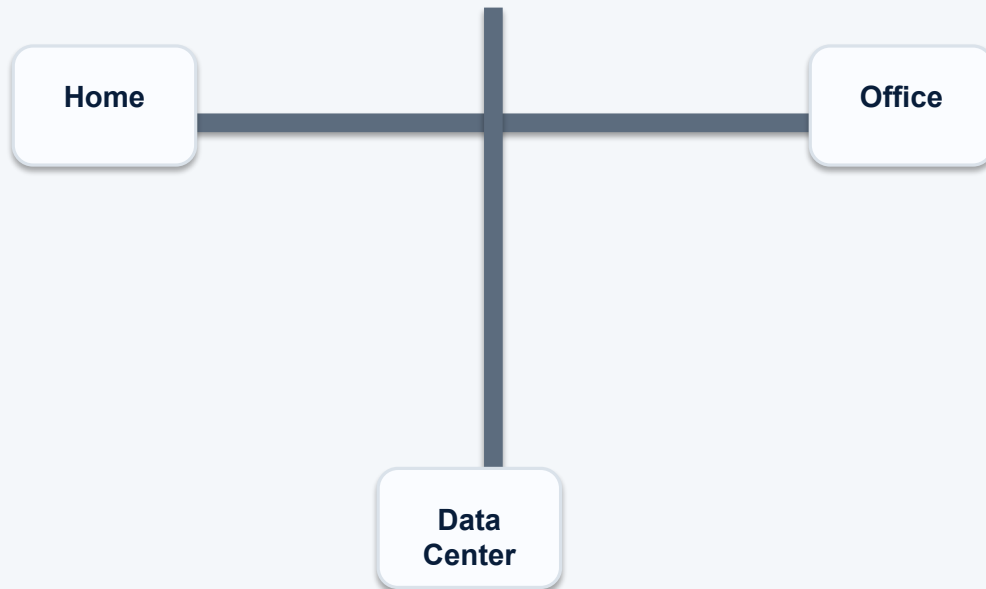
WEB

A service built on it

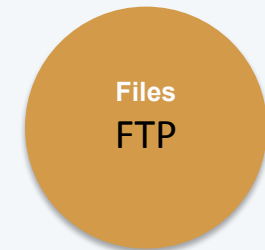
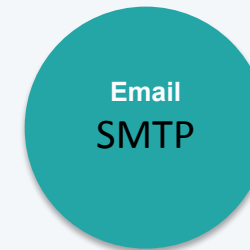
Internet vs Web: a useful analogy

Think of infrastructure first, then the service that uses it.

INTERNET = ROAD NETWORK



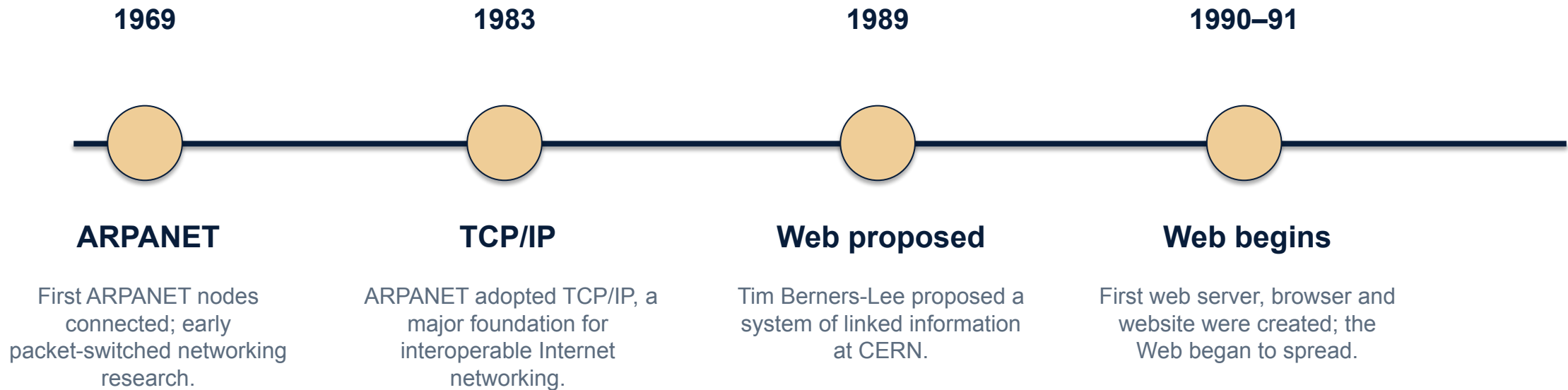
WEB = ONE SERVICE USING THE ROADS



The Internet can exist without the Web. Email, file transfer, voice calls and many other services also use Internet infrastructure.

A short history: from ARPANET to the Web

Only the milestones needed to understand how today's Web emerged.



Key idea: the Internet is older than the World Wide Web.

What is a protocol?

Communication works only when both sides follow agreed rules.

Definition

A network protocol is a defined set of rules for formatting, transmitting and receiving data between systems. Protocols specify how communicating devices should interpret messages and respond to them.

Analogy: a shared language

Two people can communicate effectively only when they understand the same language and rules.
Computers need the same agreement.

HTTP / HTTPS

Web communication

SMTP

Sending email

FTP

File transfer

DNS

Name resolution

SMTP and FTP

Two classic application-layer protocols with very different jobs.

SMTP — Simple Mail Transfer Protocol

SMTP is primarily used to send and relay email between mail clients and mail servers, and between mail servers. Receiving email is typically handled by other protocols such as IMAP or POP3.



FTP — File Transfer Protocol

FTP is designed to transfer files between a client and a server. Traditional FTP does not encrypt traffic, so secure alternatives such as SFTP or FTPS are preferred when protection is required.



HTTP and HTTPS

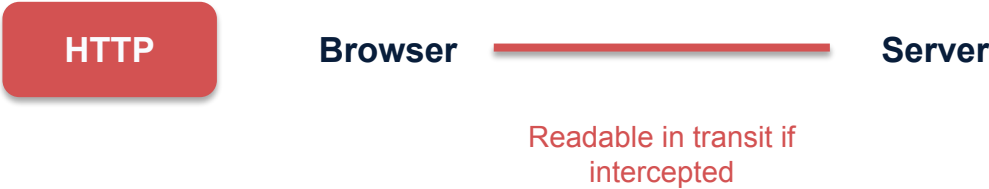
The primary protocols used for communication on the Web.

HTTP — Hypertext Transfer Protocol

HTTP is an application-layer protocol that defines how web clients and servers exchange requests and responses. It is used to transfer HTML, JSON, images and many other web resources.

HTTPS — HTTP over TLS

HTTPS is HTTP carried over a TLS-secured connection. TLS provides encryption, integrity protection and server authentication, which helps protect data while it travels between browser and server.



Anatomy of a URL

A URL tells the browser where a resource is and how to access it.

`https://www.example.com:443/products/shoes?id=42#reviews`

Scheme

`https://`

How to access the resource

Host

`www.example.com`

Which host/domain

Port

`:443`

Network service endpoint

Path

`/products/shoes`

Resource path

Query

`?id=42`

Extra parameters

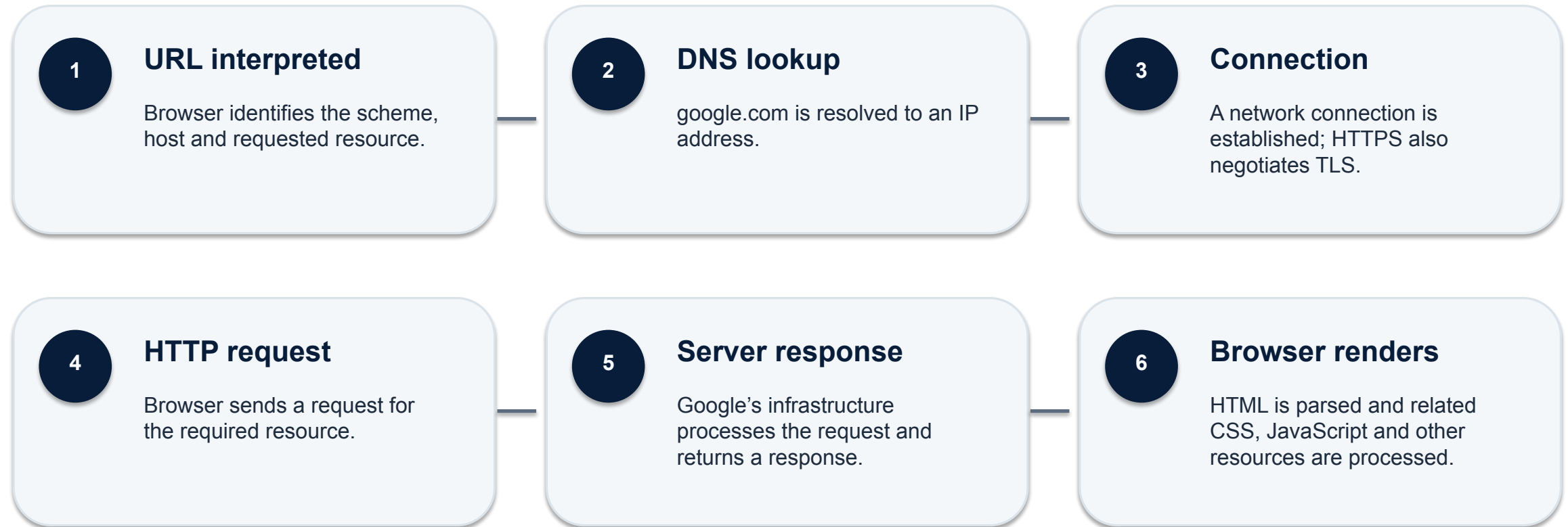
Fragment

`#reviews`

Location inside the page

What happens when you enter google.com?

A simplified end-to-end view of the browser journey.



DNS and IP addresses

Names are convenient for people; IP addresses are used for network routing.

Domain Name

A domain name is a human-readable name such as google.com. It is easier to remember than a numerical network address and can be mapped to one or more IP addresses through DNS.

DNS

The Domain Name System is a distributed naming system. It translates domain names into records that help clients locate services, commonly including IPv4 or IPv6 addresses.

IP Address

An IP address identifies a network interface for Internet Protocol communication and routing. IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses.



HTTP request and response

The client asks; the server answers.



WEB DEVELOPMENT TOOLS

The software and utilities that support the development workflow.

Node.js

Package Managers

VS Code

Web Server

Node.js

A JavaScript runtime used beyond the browser.

Definition

Node.js is an open-source, cross-platform JavaScript runtime built on the V8 JavaScript engine. It allows JavaScript to run outside a web browser, including on servers and in command-line tools.

Why web developers use it

Node.js is commonly used to build backend services, APIs, development tools and automation scripts. Modern frontend tooling also relies heavily on the Node.js ecosystem.



Examples: Express servers • build tools • scripts • package tooling

Package managers

Managing reusable code and project dependencies.

Definition

A package manager is a tool that installs, updates, removes and tracks software packages used by a project. In the JavaScript ecosystem, packages are commonly distributed through registries such as npm.

Common choices

npm is distributed with Node.js. Other widely used JavaScript package managers include Yarn and pnpm. A project should normally use one package manager consistently.

package.json

Declares project metadata and dependencies

Package Manager

Resolves and downloads packages

node_modules

Installed dependency files

Lock file

Records exact resolved versions

Installing Node.js

A clean setup for a development machine.

1

Download

Use the official Node.js distribution for your operating system. For most learners, choose the current LTS release.

2

Install

Run the installer and keep npm enabled. The installer adds the Node.js and npm commands to your system.

3

Verify

Open a terminal and run `node --version` and `npm --version` to confirm that both commands are available.

4

Create a project

Create a folder, open it in VS Code, and initialize a project when needed with `npm init`.

Verification commands: `node --version` `npm --version`

Visual Studio Code

A lightweight source-code editor with a large extension ecosystem.

What it is

Visual Studio Code is a code editor used for writing, navigating, debugging and managing source code. It supports many languages and can be extended with additional tooling.

Useful built-in capabilities

Integrated terminal, Git support, search, debugging, code navigation, syntax highlighting and workspace management make it suitable for modern web development.

EXPLORER

index.html
style.css
script.js

EDITOR

`<h1>Hello Web</h1>`

TERMINAL

`npm run dev`

Extensions in VS Code

Extensions add language support, formatting, linting and workflow features.

Definition

A VS Code extension is an installable add-on that extends the editor with new capabilities. Extensions can provide language intelligence, formatting, themes, debugging support and integrations.

Use them selectively

Install extensions that solve a real need. Too many overlapping extensions can slow the editor or create conflicting formatting and linting behavior.

Prettier

Code formatting

ESLint

JavaScript/TypeScript linting

Live Server

Simple local static server

GitLens

Additional Git context

Web server

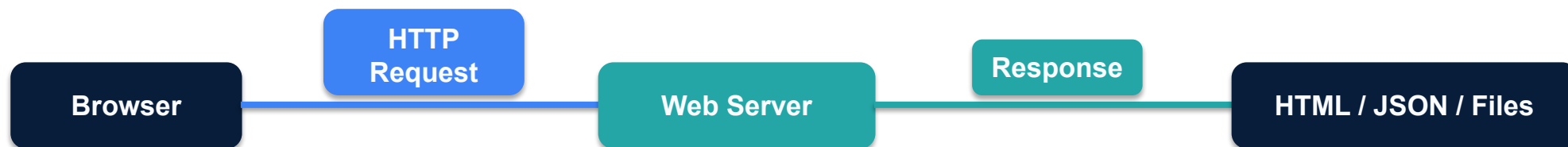
The component that receives web requests and returns web resources or application responses.

Definition

A web server is software that listens for HTTP/HTTPS requests and returns responses. It may serve static files directly, forward requests to application code, or perform both roles.

Examples

Nginx and Apache HTTP Server are common web servers. Node.js applications can also create HTTP servers directly or through frameworks such as Express.



Website vs Web Page

A website is a collection; a web page is one document or view within it.

Website

A website is a collection of related web pages and resources published under a common domain or identity. A website can contain many sections, pages, images, scripts and services.

Web Page

A web page is an individual document or application view that can be accessed through a URL and displayed by a browser. Pages are commonly structured with HTML and styled with CSS.



Project folder structure

A predictable structure makes a project easier to understand and maintain.

my-website/

```
├── index.html
├── pages/
│   └── about.html
├── css/
│   └── style.css
├── js/
│   └── script.js
└── assets/
    ├── images/
    └── icons/
```

Why structure matters

A clear folder structure separates content, styles, scripts and assets. This makes files easier to locate, reduces naming confusion and helps teams follow the same project conventions.

Important principle

There is no single folder structure that fits every project. The structure should match the size and architecture of the application and remain consistent within the team.

Static pages vs Dynamic pages

The difference is about how content is produced and changed.

Static Page

A static page delivers pre-created content that is generally the same until its files are changed or rebuilt. Static pages can still use JavaScript for interaction; “static” does not mean “non-interactive.”

Dynamic Page

A dynamic page can generate or change content based on data, user identity, time, database records or application state. The content may be produced on the server, in the browser, or through both.

Portfolio / Documentation

Often mostly static

Increasing dependence on changing data

Dashboard / E-commerce

Often highly dynamic

Page composition

A web page is usually a combination of structure, presentation and behavior.



HTML

Structure & meaning

Defines headings, paragraphs, forms, links, sections and semantic structure.



CSS

Presentation

Controls layout, spacing, typography, colors, responsiveness and visual appearance.



JavaScript

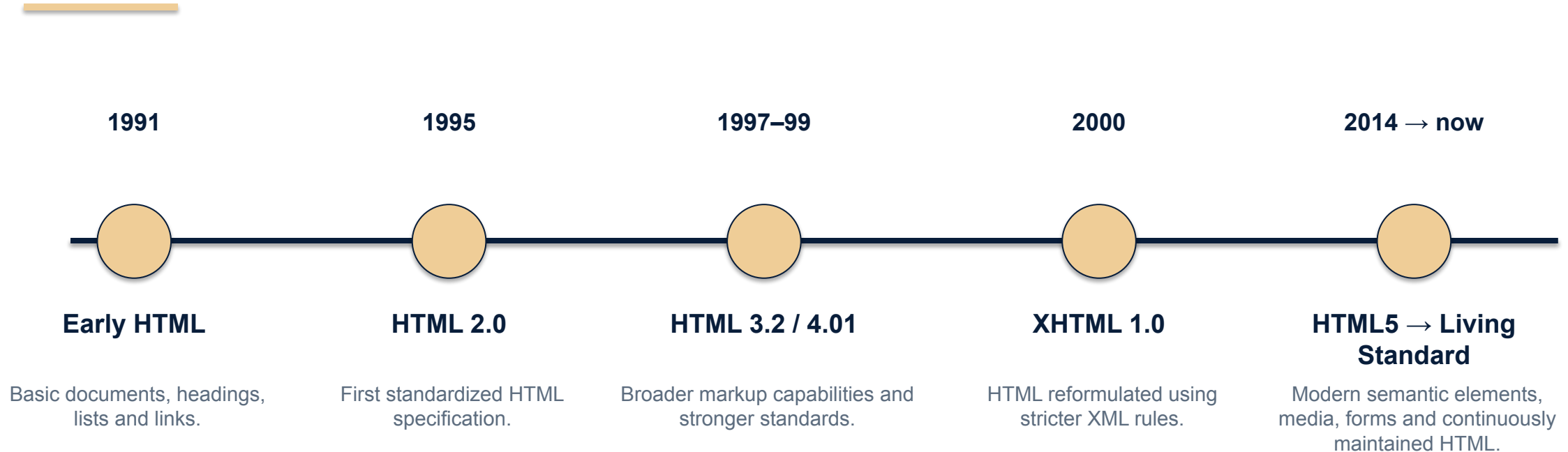
Behavior

Adds logic, interactivity, events, data fetching and updates to the page.

HTML answers “what is it?”, CSS answers “how should it look?”, JavaScript answers “how should it behave?”

Evolution of HTML

HTML evolved from a small document language into the foundation of modern web content.



Modern HTML is maintained as a Living Standard rather than waiting for a single numbered “next version.”

Putting everything together

From writing code to serving a page over the Web.



The Web development workflow connects code, tools, servers, protocols, networks and browsers into one system.

Quick recap

Use these questions to check whether the core ideas are clear.

1

How is the Internet different from the World Wide Web?

2

What role did ARPANET play in Internet history?

3

Why do computers need protocols? Give examples.

4

What happens after a browser receives a domain name?

5

What is the difference between a website and a web page?

6

Why do web developers use Node.js and package managers?

7

How are static and dynamic pages different?

8

What roles do HTML, CSS and JavaScript play in a page?



THANK YOU

Web & Internet Fundamentals

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